

A Cosmological Formula for the MOND Acceleration Scale Tested Against 3,389 SPARC Rotation-Curve Measurements

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Abstract

The MOND acceleration scale $a_0 \sim 1.2 \times 10^{-10} \text{ m/s}^2$ has been a free parameter for 42 years. Previous cosmological formulas for a_0 have residuals of 9–446%. We report the empirical relation $a_0 = c H_0 (\Omega_\Lambda / \Omega_m)^{1/6} / (2\pi)$, which predicts 1.203×10^{-10} from DESI DR2 parameters. Tested against 3,389 data points from 175 SPARC galaxies, this formula is decisively better than the previous best cosmological formula $c H_0 / (2\pi)$ ($\Delta\chi^2 = 4,895$) and statistically indistinguishable from the canonical empirical value $a_0 = 1.20$ ($\Delta\chi^2 = 38$, or 0.04% of total χ^2 , sign dependent on mass-to-light ratio). For the first time, a cosmological formula for a_0 is close enough to the empirical value that rotation-curve data cannot distinguish them. Both the 2π and the $1/6$ exponent are free parameters. The formula predicts a_0 increases by 74% at $z = 2$, distinguishable from constant MOND (0%), Λ CDM simulations (160%), and Verlinde-type scaling (200%), testable with SKA and JWST.

1. Key Results

1.1 The Formula

$$a_0 = c H_0 (\Omega_\Lambda / \Omega_m)^{1/6} / (2\pi)$$

With DESI DR2 ($H_0 = 68.17 \text{ km/s/Mpc}$, $\Omega_\Lambda = 0.6889$, $\Omega_m = 0.3111$): $a_0 = 1.203 \times 10^{-10} \text{ m/s}^2$. With Planck 2018 ($H_0 = 67.36$): $a_0 = 1.189 \times 10^{-10} \text{ m/s}^2$. H_0 for exact match to $a_0 = 1.20$: 67.98 km/s/Mpc (within 1-sigma of DESI DR2).

1.2 Improvement Over Previous Cosmological Formulas

Formula	Source	a_0 (10^{-10} m/s^2)	Residual
$c H_0$	Verlinde 2016	6.55	+446%
$c^2 \sqrt{\Lambda/3}$	Milgrom 1999	5.44	+353%
$c H_0 / 6$	Milgrom 1983	1.10	−8.3%
$c H_0 / (2\pi)$	Famaey & McGaugh 2012	1.05	−12.5%
This work		1.203	+0.29%

1.3 SPARC Rotation-Curve Test

We test against the Radial Acceleration Relation using the full SPARC database (Lelli, McGaugh & Schombert 2016): 3,389 data points from 175 galaxies. The RAR: $g_{\text{obs}} = g_{\text{bar}} / [1 - \exp(-\sqrt{(g_{\text{bar}}/a_0)})]$, evaluated with $M/L_{\text{disk}} = 0.5$, $M/L_{\text{bulge}} = 0.7$ at 3.6 micron.

Formula	a_0 (10 ⁻¹⁰ m/s ²)	χ^2	$\Delta\chi^2$ from best
Free best fit	1.245	92,923	--
This work (DESI)	1.203	93,138	+215
McGaugh 2016	1.200	93,176	+253
$cH_0/6$	1.104	95,609	+2,686
$cH_0/(2\pi)$	1.054	98,033	+5,110

The primary result: our formula beats the previous best cosmological formula $cH_0/(2\pi)$ by $\Delta\chi^2 = 4,895$. This improvement is robust across all tested mass-to-light ratios ($M/L = 0.3$ to 0.8) and error floor choices.

The comparison to McGaugh's canonical $a_0 = 1.20$ is more nuanced. The $\Delta\chi^2 = 38$ difference is 0.04% of total χ^2 (93,138). At the standard $M/L_{\text{disk}} = 0.5$, our formula gives marginally lower χ^2 . At $M/L \geq 0.6$, the sign flips and McGaugh's value gives marginally lower χ^2 . In sensitivity testing across 24 combinations of M/L ratio and error floor, our formula is better in 50% of cases. The honest characterization: our cosmological prediction is statistically indistinguishable from the empirical value.

The χ^2/N at the best fit is 27.4, far above 1.0. This reflects intrinsic scatter in the RAR that no single a_0 can remove.

1.4 Per-Galaxy Distribution

Fitting a_0 individually to 171 galaxies with 5+ data points yields: median = 0.95, mean = 1.30, std = 1.11, IQR = [0.60, 1.54] (all $\times 10^{-10}$ m/s²). Our formula (1.203) sits within the interquartile range. The wide scatter confirms a_0 is not perfectly universal across galaxies (Rodrigues et al. 2018).

1.5 Testable Redshift Prediction

$$a_0(z) = c H(z) [\Omega_\Lambda(z)/\Omega_m(z)]^{1/6} / (2\pi)$$

At $z = 2$: This work predicts 1.74 $\times$, constant MOND 1.00 $\times$, Λ CDM simulation (Mayer+ 2022) 2.60 $\times$, Verlinde $H(z)$ scaling 3.01 $\times$. All four are well separated at $z > 1$ and below Milgrom's (2017) upper bound of $\sim 4\times$ at $z \sim 2$. Testable with SKA HI and JWST H-alpha surveys at $z \sim 1-2$.

2. Free Parameters

2.1 The 2π

The Gibbons–Hawking temperature $T_{\text{dS}} = \hbar H_0/(2\pi k_B)$ contains a 2π . However, inverting via the Unruh relation $T = \hbar a/(2\pi c k_B)$ gives $a = cH_0$ — the 2π cancels. The 2π in our formula is a free parameter, not derived from horizon thermodynamics. Famaey & McGaugh (2012) noted $a_0 \sim cH_0/(2\pi)$ without claiming a derivation; we inherit their approximation.

2.2 The 1/6 Exponent

Empirical. Among integer-denominator exponents (Planck H_0): 1/5 gives 1.8% residual, 1/6 gives 0.85%, 1/7 gives 2.7%. The exponent 1/6 is the best match but 1/5 is within the 1.7% statistical uncertainty on a_0 .

2.3 Look-Elsewhere Effect

Scanning ~2,160 two-parameter formulas of the form $cH_0 \times (\Omega_\Lambda/\Omega_m)^{p/q} / (n \times \pi^m)$, 14 match a_0 within 1%. Our formula is reported for simplest algebraic form, most natural dimensionless ratio, and robust performance on the SPARC χ^2 test.

3. Sensitivity Analysis

The $\Delta\chi^2 = 4,895$ improvement over $cH_0/(2\pi)$ is robust: it holds across all tested M/L ratios (0.3 to 0.8) and error floors (0.001 to 0.1 in log space).

The $\Delta\chi^2 = 38$ comparison to McGaugh is fragile: it depends on M/L ratio. At $M/L_{\text{disk}} = 0.5$ (standard), our formula is marginally better. At $M/L_{\text{disk}} = 0.6$ or higher, McGaugh is marginally better. The two values produce statistically indistinguishable fits.

The systematic uncertainty on a_0 is 20% (McGaugh et al. 2016). However, the χ^2 test uses individual data point errors, not the global a_0 uncertainty. The $\Delta\chi^2 = 4,895$ improvement over $cH_0/(2\pi)$ reflects a genuine difference in RAR fit quality.

4. Claims and Non-Claims

Claims:

- (1) Closest published cosmological match to a_0 (0.29% with DESI DR2).
- (2) Decisively better than all previous cosmological formulas on SPARC RAR data ($\Delta\chi^2 = 4,895$ vs $cH_0/(2\pi)$; $\Delta\chi^2 = 2,471$ vs $cH_0/6$).
- (3) Statistically indistinguishable from the canonical empirical a_0 on SPARC RAR data ($\Delta\chi^2 = 38$, 0.04% of total χ^2 , sign M/L-dependent).
- (4) Testable redshift prediction distinct from all alternatives at $z > 1$.

Does not claim:

- (1) First-principles derivation. Both 2π and $1/6$ are free.
- (2) That our formula beats the empirical a_0 . The $\Delta\chi^2 = 38$ is noise-level and M/L-dependent.
- (3) Unique selection by data. 14 other formulas match within 1%.
- (4) High- z confirmation. Only upper bounds exist.

5. Conclusion

The correction $(\Omega_\Lambda/\Omega_m)^{1/6}$ applied to $a_0 \sim cH_0/(2\pi)$ reduces the gap between cosmological prediction and empirical measurement from 12.5% to 0.29%. Tested against 3,389 SPARC rotation-curve measurements, this cosmological a_0 is decisively better than all previous cosmological formulas ($\Delta\chi^2 = 4,895$ vs $cH_0/(2\pi)$) and statistically indistinguishable from the empirically measured value. The physical origin of both free parameters remains unknown.

Data Availability

SPARC data: Lelli+ 2016, CDS Strasbourg J/AJ/152/157. Cosmological parameters: Planck 2020, DESI 2025.

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